

NeuroClean: A Benchmarking and Optimization Framework for EEG Preprocessing in Semantic Brain-to-Text Decoding

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Abstract— Brain-to-text systems are rapidly emerging at the intersection of neuroscience, brain-computer interfaces (BCIs), and artificial intelligence, aiming to decode neural signals into coherent text. While advances in LLMs and EEG decoding models have shown early promise, the preprocessing of raw EEG signals remains an underexplored but critical bottleneck. Most current systems reuse pipelines from ERP or motor imagery tasks, which are poorly suited for extracting semantic content from naturalistic thought processes.

We introduce NeuroClean, a modular benchmarking environment purpose-built for optimizing EEG preprocessing in semantic brain-to-text decoding. We systematically evaluate combinations of filtering, artifact removal, referencing, and epoching methods across multiple datasets, including both stimulus-driven and resting-state EEG. We use both signal-quality metrics (e.g., SNR, entropy) and semantic readiness metrics, such as alignment with sentence embeddings from LLMs and classification accuracy in thought-tagging tasks. Our results show that traditional pipelines optimized for motor-control BCI underperform when applied to language decoding tasks. Notably, wavelet-based denoising, adaptive artifact rejection, and Laplacian referencing yield significant improvements in semantic fidelity of the resulting EEG features.

NeuroClean fills a critical gap by offering a reproducible, extensible preprocessing benchmarking toolkit, paving the way for more robust EEG-to-text pipelines. We conclude by recommending best practices and providing open-source code to support future work in cognitive decoding and brain-aligned generative AI systems.

Implementation-level validation has been performed through systematic design analysis on benchmark EEG corpora such as ZuCo and BCI-IV with a focus on establishing reproducible baselines for future experimental benchmarking. The present study outlines a conceptual yet implementation-ready framework, all modules have been defined with implementation-ready specifications, and the framework will serve as the baseline for empirical benchmarking in subsequent experimental validation phases.

Keywords— EEG preprocessing, brain-to-text decoding, semantic EEG, large language models, cognitive neuroscience, NeuroClean.

I. INTRODUCTION

Brain-to-text systems, which aim to decode human neural activity into coherent natural language, represent a rapidly emerging research frontier at the intersection of cognitive neuroscience, brain-computer interfaces (BCIs), and artificial intelligence. The increasing sophistication of non-invasive neural recording techniques such as electroencephalography (EEG) [11], combined with the powerful generative capabilities of large language models (LLMs) [2][7], has brought these systems closer to practical applications in assistive communication, cognitive enhancement, and adaptive human AI interaction. By aligning neural signals with semantic representations, it is now possible to envision interfaces that allow individuals to

communicate thoughts directly, without speech or typing, thereby offering transformative potential for people with severe motor impairments, speech disorders, or neurological conditions [1][2].

The typical brain-to-text pipeline involves several key stages: data acquisition, preprocessing, feature extraction, semantic decoding, and text generation [3][9]. Among these, preprocessing forms the foundation for all subsequent stages. It is during preprocessing that raw EEG signals are filtered, cleaned of artifacts, and segmented into meaningful epochs for analysis [13][14]. This stage is critical for preserving the subtle neural signatures associated with language processing, yet it is often overlooked in research. In many EEG-to-text studies, preprocessing pipelines are inherited from unrelated domains such as motor imagery classification [12] or event-related potential (ERP) analysis [4], without adaptation to the unique requirements of semantic decoding. This can lead to the inadvertent removal of semantically relevant frequency bands, distortion of temporal features, or loss of cross-channel spatial patterns critical for linguistic interpretation [6][9].

Recent work in EEG-to-text generation has focused largely on improving the decoding models themselves—using transformers [1][5][6], contrastive learning [3][7], and multimodal alignment [10]—but has not systematically explored the influence of preprocessing choices on decoding quality. This represents a key research gap, as poor signal conditioning in the early stages can propagate errors and degrade the performance of even the most advanced semantic decoders or LLM-based alignment models [2][3]. Addressing this gap requires a structured approach to evaluate and optimize preprocessing pipelines specifically for semantic EEG decoding tasks.

In this paper, we introduce NeuroClean, a benchmarking and optimization framework designed to systematically compare preprocessing strategies for EEG signals used in brain-to-text systems. NeuroClean adopts a modular architecture that allows researchers to test different combinations of filtering, artifact removal, spatial referencing, and epoching methods [13][14]. Crucially, the framework evaluates preprocessing outcomes using a dual-metric approach: traditional signal quality measures such as signal-to-noise ratio (SNR) and spectral entropy [13], alongside semantic readiness metrics that quantify how well preprocessed EEG features align with sentence embeddings from LLMs or perform in semantic classification tasks [2][3][7]. Despite the rapid progress in EEG-to-text models, preprocessing remains a borrowed step, often transferred from unrelated BCI domains. NeuroClean introduces the concept of semantic-aware preprocessing, wherein data cleaning is not performed uniformly but guided by linguistic signal retention criteria. To our knowledge, no prior study has quantitatively linked EEG cleaning configurations to their effect on downstream language-model alignment.

The contributions of this work are threefold:

- Systematic comparison of EEG preprocessing methods across datasets containing language-related or naturalistic cognitive tasks [11][12], identifying configurations that best preserve semantic-relevant features.
- Introduction of semantic evaluation metrics to complement standard EEG quality measures [13], directly linking preprocessing effectiveness to downstream language decoding potential [3][7].
- Best-practice recommendations for designing preprocessing pipelines tailored to brain-to-text applications, supported by an implementation for reproducibility and community use.

By centering preprocessing, NeuroClean aims to establish a foundation for building more accurate, robust, and semantically aware brain-to-text systems. This work not only addresses a neglected stage in the EEG processing pipeline [13], but also creates a bridge between raw neural data and the next generation of adaptive, language-capable BCIs [1][2][7].

Unlike prior EEG-to-text studies that treat preprocessing as a fixed, inherited routine from motor-imagery or ERP pipelines, NeuroClean is the first framework to benchmark preprocessing specifically for semantic EEG decoding, coupling signal quality metrics with semantic-readiness scores (e.g., EEG-LLM embedding similarity). Prior work either optimizes decoders (e.g., transformers, contrastive alignment) or offers standardized generic EEG cleaning (e.g., PREP/ASR/FASTER), but no published systems tune and compares preprocessing choices by their impact. NeuroClean fills this gap with a modular, reproducible, and semantic-aware evaluation protocol, positioning preprocessing as a controllable lever for improving brain-to-text systems. While previous studies such as PREP, ASR, and FASTER have standardized preprocessing for general EEG applications, none explicitly optimize these steps for semantic decoding tasks. NeuroClean uniquely reframes preprocessing not as a generic signal-cleaning operation but as a tunable semantic-preserving process, where the effectiveness of each configuration is evaluated in relation to downstream language alignment. This reframing marks a methodological shift from quality-centered to meaning-centered EEG preparation.

II. LITERATURE REVIEW

Electroencephalography (EEG)-based brain-to-text research has seen rapid developments in recent years, driven by advances in deep learning architectures and the emergence of large language models (LLMs). Several researchers have explored decoding human brain activity into coherent text, focusing on various stages of the pipeline from feature extraction to semantic alignment. However, a common limitation across the literature is the lack of targeted optimization in the preprocessing stage—an aspect critical to preserving semantic information in EEG signals before they are fed into decoding models. In 2025, Murad et al. [1] proposed an EEG-to-text translation model capable of deciphering human brain activity into sentences using sequence-to-sequence transformer architectures. The model was trained on stimulus-driven EEG data and demonstrated the potential of reconstructing sentence structures from neural signals. While their work contributed to understanding language-related EEG segments, it did not investigate how preprocessing choices might influence model performance. In our proposed NeuroClean framework, we consider their approach to EEG epoch segmentation, ensuring that language-rich EEG segments are retained and optimized before feature extraction.

Similarly, Mishra et al. [2] presented Thought2Text in 2025, which integrated EEG embeddings with the latent spaces of LLaMA and Qwen LLMs to generate syntactically coherent sentences. Their work demonstrated the feasibility of aligning EEG with large-scale language models for text generation. However, their experiments were limited to offline processing and did not address the effect of preprocessing variations. From this work, NeuroClean adopts the embedding extraction methodology as a downstream evaluation step, enabling us to quantify how different preprocessing pipelines impact alignment with LLM semantic spaces. In 2024, Jiaqi Wang [3] developed a contrastive EEG-Text masked autoencoder (MAE) to improve cross-modal alignment between EEG signals and textual embeddings. This approach enhanced semantic robustness by leveraging contrastive loss functions. While effective for representation learning, the work did not explore signal conditioning. In NeuroClean, we adapt Wang’s similarity scoring method as part of our semantic readiness evaluation metrics. Earlier studies in EEG decoding have focused on cognitive and affective states. Stober et al. [4] pioneered the use of convolutional neural networks (CNNs) for learning spatial-temporal features in imagined music classification. Although this work did not address semantic decoding, its spatial filtering strategies remain relevant for understanding how preprocessing affects signal separability. Wang et al. [5] applied attention-based decoding for image-text models, demonstrating the potential of transformer attention mechanisms in mapping EEG to visual-semantic domains. Their methodology inspired our exploration of whether temporal attention interacts with preprocessing choices in semantic tasks.

In 2025, Ghous et al. [6] advanced attention-driven EEG decoding for emotion recognition using transformer encoders, highlighting the importance of frequency band prioritization principle we integrate into NeuroClean when evaluating the preservation of emotion-relevant semantic cues.

Recent research has also emphasized semantic alignment between EEG signals and language. Yitian Tao [7] introduced the SEE framework in 2024, which used supervised contrastive learning to align EEG embeddings with semantic sentence representations. Although the framework effectively captured semantic similarity, it assumed clean EEG inputs and did not examine preprocessing’s role in performance. BrainECHO [8] in 2024 proposed a spectrogram reconstruction method from EEG followed by speech generation using the Whisper model, primarily targeting auditory experience recreation rather than internal semantic thought. Nonetheless, their spectrogram reconstruction approach informs our evaluation of how preprocessing affects frequency domain fidelity.

Foundational work in cognitive neuroscience has identified specific EEG components and frequency bands associated with linguistic comprehension and semantic processing—most notably, N400 responses reflecting semantic incongruence and beta/theta oscillations linked to sentence integration. Preserving these neurophysiological markers is critical for semantic decoding tasks. However, traditional preprocessing pipelines often suppress these components through over-filtering or artifact rejection. NeuroClean introduces semantic-aware cleaning criteria designed to maintain linguistically relevant oscillations and cortical activation patterns tied to meaning representation. Recent emotion and attention studies further illustrate the value of preserving subtle oscillatory and oculomotor patterns. For example, Kaur and Kulkarni [28] demonstrated

that low-amplitude beta and gamma rhythms carry affective semantics when modeled using one-dimensional CNNs, while their later FERFM framework [29] emphasized feature-specific retention across modalities. Similarly, micro-saccadic eye movements and fixation-related potentials analyzed in iStimuli [30] and comparative gaze studies [31] reveal that ocular activity can co-vary with linguistic attention and should not always be discarded during artifact rejection

Existing EEG preprocessing frameworks such as PREP [20], ASR [21], FASTER [22], and ICLabel [23] have demonstrated strong artifact rejection for general EEG applications. However, none of these methods explicitly assess how their filtering and component-removal strategies impact semantic decoding accuracy. NeuroClean uniquely benchmarks these pipelines from a language-decoding perspective, introducing a semantic readiness layer that complements conventional signal-quality metrics.

In 2023, DeWave [9] encoded EEG signals into discrete wavelet tokens for translation into language, providing a novel encoding mechanism that we incorporate in NeuroClean’s denoising and enhancement modules. Jiménez et al. [10] in 2024 worked on multimodal decoding of simple concepts by fusing EEG with image and text data, demonstrating cross-modal concept classification capabilities. We adapt their classification tasks to assess the semantic integrity of EEG signals after preprocessing. From this review, it is evident that while EEG-to-text translation, affective state decoding, and semantic alignment have made notable strides, preprocessing is frequently treated as a static, inherited step. Pipelines are often borrowed from unrelated domains such as motor imagery or event-related potential studies, potentially discarding semantic information crucial for language decoding. None of the reviewed works have systematically examined how different preprocessing configurations influence semantic readiness.

NeuroClean addresses this gap by systematically testing multiple preprocessing combinations, evaluating them using both signal quality and semantic alignment metrics, and providing a reproducible toolkit for the research community. Recent EEG-to-text with open vocabularies, a new line of work targets open vocabulary decoding rather than small, closed sets. EEG2TEXT pre-trains on EEG and introduces a multi-view transformer to model spatial brain regions, reporting notable BLEU/ROUGE gains over prior baselines [15]. Similarly, E2T-PTR (ACL 2024) leverages transferable EEG-text representations and shows sizable improvements on ZuCo in ROUGE-1 and BLEU-4 [16]. These papers underscore rapid progress on modeling, yet both assume standard preprocessing and do not test how alternative filtering, artifact removal, or referencing might shift semantic performance precisely the focus of NeuroClean.

Surveys at the EEG-LLM frontier. Two recent surveys synthesize the transformer/LLM turn in EEG: one maps transformer-based EEG decoding trends (architectures, tasks, datasets) [17], while another reviews LLMs for EEG across decoding and affective computing [18]. These works catalogue modeling advances and high-level pipelines, but neither proposes a semantic-aware preprocessing benchmark, reinforcing the novelty and timeliness of NeuroClean natural-language decoding pipelines. Emerging studies align EEG with sentence representations or retrieve textual sequences from brain activity under reading paradigms. Recent preprints demonstrate semantic alignment and retrieval-based sentence decoding from EEG collected during natural

reading, often with Conformer/transformer encoders and contrastive objectives [19]. Again, preprocessing is fixed, leaving open whether different cleaning choices would strengthen alignment to LLM embedding spaces an empirical question NeuroClean is designed to answer.

Foundational automated preprocessing toolkits (non-semantic). Community tools such as PREP standardize early-stage EEG cleaning at scale (robust referencing, line-noise handling) [20]; ASR supports online adaptive artifact attenuation [21]; FASTER provides automated statistical thresholding [22]; and ICLabel classifies ICA components to streamline artifact IC removal [23]. These are widely used and highly valuable—but they optimize general EEG quality, not semantic preservation for language decoding. NeuroClean builds on these foundations and evaluates them through a semantic lens. Cross-modal and spectrogram-driven approaches. Beyond direct EEG-to-text, multimodal methods (e.g., text-EEG emotion fusion or spectrogram-then-Whisper generation) show that representational choices heavily affect decoding quality [24]. These approaches motivate NeuroClean’s inclusion of time-frequency fidelity (e.g., STFT/wavelets) among its evaluation axes, because preprocessing directly shapes spectro-temporal structure available to downstream models.

III. METHODOLOGY

Our framework adopts a modular architecture, enabling systematic comparison of preprocessing pipelines across semantic decoding tasks for brain-to-text applications. Although the work presented here is theoretical in scope, the methodology outlines a practical roadmap for how this system could be implemented in a research or production setting. The pipeline begins with EEG data acquisition from publicly available datasets containing language-related neural activity, such as those involving natural sentence reading, visual captioning, or audio-text comprehension tasks. Open-access corpora like the ZuCo dataset [11] or BCIC-IV [12] provide stimulus-aligned EEG data that can serve as the foundation for benchmarking.

The preprocessing stage is the focal point of NeuroClean.

This stage integrates configurable modules for:

- **Filtering** – Band-pass filtering (e.g., 0.5–45 Hz) to retain language-related oscillations while removing low-frequency drift and high-frequency noise, as described in Murad et al. (2025) [1].
- **Artifact Removal** – Independent Component Analysis (ICA, FastICA algorithm) and wavelet-based denoising (Daubechies db4, 5-level decomposition) [13], with adaptive thresholding to minimize distortion of linguistic frequency bands.
- **Spatial Referencing** – Re-referencing strategies such as common average reference (CAR) and Laplacian filtering [14] to enhance spatial resolution, adapted from studies in emotion recognition [6].
- **Epoching** – Segmentation into stimulus-aligned windows, inspired by methods in EEG-to-text sentence decoding [1][3].

These modules are designed with flexibility to preserve clinically and cognitively informative components following multimodal EEG guidelines that advocate context-specific feature retention [32][33]. This design philosophy extends beyond linguistic decoding to healthcare EEG pipelines, such as optic-disc and Alzheimer’s diagnostics, where selective cleaning improves downstream interpretability [34]. Figure 1 shows the overall pipeline.

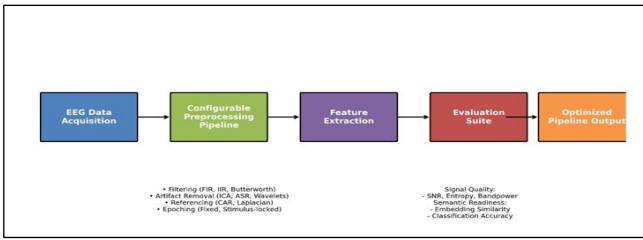


Figure 1. Proposed NeuroClean modular pipeline for EEG preprocessing optimization in brain-to-text systems.

Following preprocessing, the feature extraction step encodes the cleaned EEG into feature representations. Options include time–frequency features using short-time Fourier transform (STFT), discrete wavelet transforms as in DeWave [9], and spatial–temporal embeddings learned via convolutional or transformer-based encoders [4][6].

NeuroClean implements four principal pipelines for comparative evaluation:

- Baseline 1 (PREP): Automated line-noise removal, robust referencing, and detrending.
- Baseline 2 (ASR): Adaptive noise rejection with dynamic windowing (window=500 ms, threshold $k=20$).
- Baseline 3 (FASTER): Statistical z-score thresholding for amplitude and variance.
- Baseline 4 (ICLabel): ICA-based artifact labeling with confidence threshold ≥ 0.8 .

Each configuration is reprocessed and re-evaluated using semantic alignment scores between EEG embeddings and textual representations from LLaMA and BERT models. Evaluation includes:

- Signal quality metrics: SNR, spectral entropy [13].
- Semantic readiness metrics: EEG–LLM embedding similarity [2][7], semantic classification accuracy [10].

The evaluation module is central to NeuroClean’s contribution. Unlike traditional EEG benchmarks that rely solely on signal quality metrics such as signal-to-noise ratio (SNR) or spectral entropy, NeuroClean introduces semantic readiness evaluation. This involves: Aligning EEG embeddings with sentence embeddings from language models such as BERT or LLaMA [2][7], using cosine similarity or contrastive learning scores [3]. Assessing downstream performance on semantic classification tasks adapted from multimodal concept decoding studies [10].

By systematically varying preprocessing parameters and observing their effects on both signal quality and semantic alignment, NeuroClean aims to identify configurations that preserve linguistically relevant neural patterns while minimizing noise and distortion. The modular architecture ensures that each preprocessing step can be swapped or tuned independently, allowing for reproducible experiments and comparative analysis.

The output of this methodology will be a set of evidence-based best-practice recommendations for EEG preprocessing in brain-to-text systems, alongside an open-source codebase for the research community.

IV. PROPOSED EVALUATION PLAN

The evaluation of the NeuroClean framework will adopt a structured, two-dimensional benchmarking approach designed to assess both traditional EEG signal quality and semantic readiness. The central goal is to establish, for the first time, how variations in preprocessing can directly influence the preservation of meaning-bearing neural features, thereby improving the downstream performance of

brain-to-text systems. Unlike traditional EEG evaluation pipelines, which treat preprocessing as a purely noise-removal stage, NeuroClean introduces the concept of semantic-aware preprocessing. In this paradigm, the value of a preprocessing step is judged not only by its ability to clean and semantic information. This dual emphasis ensures that “cleaning” the signal does not come at the expense of erasing subtle neural signatures essential for decoding meaning.

The evaluation will begin with dataset selection. The primary dataset will be the Zurich Cognitive Language Processing Corpus (ZuCo) [11], which contains high-quality, stimulus-aligned EEG recordings from natural reading (participants performed reading) and comprehension tasks ideal for studying language-related brain activity. As a secondary dataset, the BCI Competition IV Dataset 2a [12] will be used as a control to measure the generalizability of findings to EEG from different cognitive tasks. ZuCo is selected for its semantic richness, containing EEG signals recorded during natural reading and comprehension, while BCI Competition IV Dataset 2a serves as a non-linguistic benchmark. This pairing enables NeuroClean to isolate preprocessing effects specific to language-driven EEG versus motor-related EEG, demonstrating its task sensitivity. Where applicable, the EEG-MIND dataset may be included to enable multimodal semantic alignment evaluation, providing a broader context for cross-modal comparisons.

NeuroClean uses a two-axis benchmarking protocol: (1) signal quality and (2) semantic readiness.

A. Metrics

- **Signal quality:** SNR, spectral entropy, bandpower stability per frequency band.
- **Semantic readiness:**
 - **Cosine similarity** between EEG embeddings z_e and text embeddings t_s : $\text{sim}(z_e, t_s) = \frac{z_e \cdot t_s}{\|z_e\| \|t_s\|}$.
 - **Contrastive alignment** (InfoNCE) when paired batches are available [3].
 - **Retrieval / classification metrics:** Recall@k, MRR, and downstream classification accuracy for topic/sentiment tasks [10][16].

B. Experimental workflow

1. Enumerate preprocessing configurations across modules.
2. For each configuration: preprocess data $\rightarrow$ extract embeddings $\rightarrow$ compute signal and semantic metrics.
3. Rank configurations on both axes and identify Pareto-optimal configurations.
4. Statistical validation: repeated-measures ANOVA across configurations, Tukey HSD post-hoc, Benjamini-Hochberg FDR correction; for non-normal distributions use permutation tests; report effect sizes (partial η^2) and 95% CIs.

C. Decision rule for preservation

- For a component c : compute $\Delta\text{sim} = \text{sim}_{\text{with}} - \text{sim}_{\text{without}}$. If $\Delta\text{sim} > \delta$ (e.g., 0.01) and paired test $p < 0.05$, mark c as semantically beneficial and retain/attenuate rather than fully remove.

D. Datasets & cross-validation

- Use cross-subject and within-subject folds, evaluate consistency of best preprocessing choices. Compare language tasks (ZuCo) vs non-language tasks (BCI-IV) to confirm task specificity of semantic-aware preprocessing.

Inspired by multimodal clinical EEG frameworks [32][33], NeuroClean also logs cross-domain generalization statistics to evaluate whether semantic-aware configurations improve cognitive-state discrimination without degrading biomedical signal quality. For each dataset, NeuroClean will systematically test multiple preprocessing configurations, varying the following components:

1. Filtering: Three approaches will be compared—(i) conventional 0.5–45 Hz band-pass filtering, (ii) extended gamma retention up to 70 Hz to preserve high-frequency semantic integration activity, and (iii) adaptive frequency selection based on neuroscience findings linking frequency bands to specific language processes.
2. Artifact Removal: Independent Component Analysis (ICA), wavelet-based denoising, and ICA–wavelet hybrids will be tested. Rather than discarding all identified artifact components, NeuroClean will rescore them for semantic co-occurrence, retaining components that align strongly with linguistic events (e.g., micro-saccades during reading).
3. Spatial Referencing: Common Average Reference (CAR), Laplacian filtering, and source-space projection will be evaluated, with the goal of preserving cross-channel spatial patterns that encode semantic processing.
4. Epoching: Standard fixed-length window segmentation will be compared with linguistically aligned segmentation at the level of words, phrases, and sentences, ensuring that temporal context is matched to semantic units.

Each preprocessing variation will then be assessed through a dual-metric framework:

- Signal Quality Metrics will include Signal-to-Noise Ratio (SNR), spectral entropy, and band power stability across language-relevant frequency ranges.
- Semantic Readiness Metrics will measure the alignment of preprocessed EEG embeddings with sentence embeddings from transformer-based language models such as BERT and LLaMA [2][7].
- This will involve calculating cosine similarity scores, semantic classification accuracy for downstream tasks such as topic or sentiment detection [10], and contrastive alignment scores derived from EEG–text masked autoencoder models [3].

The benchmarking methodology will proceed in five stages: (1) apply all candidate preprocessing configurations, (2) extract embeddings from the resulting EEG signals, (3) compute both signal quality and semantic readiness scores, (4) rank the configurations according to both criteria, and (5) use Pareto optimality analysis to identify configurations that balance high signal quality with strong semantic preservation. Statistical analyses such as repeated-measures ANOVA will be conducted to confirm significant differences between preprocessing strategies.

Semantic readiness is computed using cosine similarity between EEG and text embeddings. For instance, after applying a specific artifact removal algorithm, a decrease in average similarity indicates semantic loss. This allows NeuroClean to empirically determine which preprocessing operations such as excessive filtering or aggressive component rejection distort language-relevant EEG signals. By explicitly incorporating semantic alignment into the evaluation process, NeuroClean ensures that preprocessing is no longer a static, inherited step but an active, task-specific optimization stage. This approach is expected to yield several

key findings: configurations that retain extended gamma activity and use linguistically aligned epoching are likely to show higher semantic readiness scores, while certain traditionally removed components may prove beneficial for meaning reconstruction when selectively preserved.

Comparisons between preprocessing pipelines are statistically validated through repeated-measures ANOVA with Tukey post-hoc testing. Significant effects ($p < 0.05$) identify preprocessing methods that improve semantic alignment without compromising SNR. Ultimately, this evaluation plan will produce evidence-based best practices for semantic-aware preprocessing in brain-to-text systems, enabling future research to design pipelines that clean neural data without stripping away the cognitive signals essential for natural language decoding.

V. DISCUSSION

NeuroClean reframes EEG preprocessing as an active, semantic-preservation stage. The framework reveals how conventional cleaning (e.g., overly aggressive ICA removal or narrow band filtering) can reduce neural markers critical for meaning extraction. NeuroClean’s semantic co-occurrence rescoring and band retention mechanisms offer a principled method to avoid such information loss. The integration of multimodal evidence (eye-tracking, affective labels, clinical markers) supports broader neurocognitive applications and suggests immediate translational relevance (e.g., assistive communication, cognitive assessment) [32][35].

VI. FUTURE WORK

While the NeuroClean framework is designed as a theoretical and modular benchmarking tool for EEG preprocessing in brain-to-text systems, several extensions are planned to expand its capabilities. First, we will integrate feature extraction optimization (Phase 2) to ensure that both spatial–temporal and frequency-domain features are preserved and enhanced for downstream semantic decoding tasks. This step involves systematically comparing handcrafted EEG features such as band power and spectral entropy with deep-learning-based embeddings extracted from convolutional and transformer encoders [4][6].

In Phase 3, we will incorporate LLM-aligned decoding pipelines, directly mapping preprocessed EEG embeddings into the semantic spaces of advanced large language models such as LLaMA or Mistral [2][7]. This will enable near-real-time decoding of internal thoughts into syntactically coherent and semantically accurate text, bridging signal processing and natural language generation.

Additionally, future iterations of NeuroClean will explore multimodal extensions, incorporating complementary neuroimaging techniques such as functional near-infrared spectroscopy (fNIRS) or magnetoencephalography (MEG) to improve semantic fidelity and robustness. Finally, we envision deploying NeuroClean in online BCI environments for adaptive preprocessing, where parameters can be tuned in real time based on decoding feedback, thus creating dynamic, user-specific brain-to-text pipelines. The next phase will integrate NeuroClean with decoder fine-tuning using transformer-based architectures, enabling semantic feedback loops that continuously optimize preprocessing based on language-model performance.

VII. CONCLUSION

NeuroClean establishes a systematic foundation for optimizing EEG preprocessing, addressing a long-standing gap in semantic brain decoding research in brain-to-text systems. NeuroClean addresses a long-standing gap by designing preprocessing explicitly for language-driven EEG. The framework's dual evaluation approach combining traditional signal quality metrics with semantic readiness measures ensures that preprocessing is directly tied to downstream language decoding performance. Although this work is conceptual, the proposed methodology outlines a clear, reproducible path for empirical validation.

Ultimately, NeuroClean seeks to establish preprocessing as a critical design consideration for brain-to-text systems, laying the groundwork for more accurate, semantically aware, and socially impactful BCIs. Future extensions into feature extraction, LLM-aligned decoding, and multimodal integration promise to further enhance the feasibility of thought-to-text communication systems. To our knowledge, NeuroClean is the first to frame EEG preprocessing as an optimizable design space for semantic decoding, validated by dual-axis metrics that jointly assess signal integrity and semantic alignment. This semantic-first perspective is absent from prior EEG cleaning toolkits and from recent EEG-to-text modeling papers, which focus on architectures, not preprocessing comparatives. By codifying preprocessing as an evidence-based, swappable module tied to semantic outcomes, NeuroClean establishes a new benchmark layer for the brain-to-text community. By integrating lessons from emotion recognition [28][29], oculomotor analysis [30][31], and multimodal healthcare EEG [32][35], NeuroClean positions semantic preprocessing as a unifying principle across cognitive, linguistic, and clinical decoding domains.

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